

Status and scope. The entire THM047 conjunction has passed fresh isolated reconstruction (AUD067) and hostile mathematical review (AUD068). Root has completed the full comparison, exact source qualification and sealed-evidence verification; the bounded gate is accepted. The original critical dynamic source assertion THM046 remains open. Work on the unit four-torus with coefficient-one Coulomb kernel, $K = -\nabla g$, $c = 4\pi^2$ and $\operatorname{div} K = c(\delta_0 - dx)$. The actual particles start from iid Haar, with drift $N^{-1} \sum_{j \neq i} K(X_i - X_j)$ and independent noise $\sqrt{2\nu_N} dW_i$. Here $\nu_N = \beta_N^{-1}$ and $\lambda_N = \beta_N N^{-1/2} \rightarrow \lambda \in (0, \infty)$.

For smooth real h and finite T , set $f_t = Q_{T-t}^{\nu_N} h$. The operator $Q_a^{\nu_N}$ preserves constants and has multiplier $e^{-a(c+c\nu_N|k|^2)}$ on nonzero modes. Define

$$J_t(x, y) = K(x - y) \cdot (\nabla f_t(x) - \nabla f_t(y)).$$

Let Φ_t be the genuine symmetric pair inverse with terminal value zero and both responses. Write $q_t(x) = \int \Phi_t(x, y) dy$. Its off-diagonal equation is

$$\begin{aligned} \partial_t \Phi + \nu_N \Delta_{x,y} \Phi + N^{-1} B\Phi + R\Phi &= -J, \\ B\Phi &= K(x - y) \cdot (\nabla_x - \nabla_y) \Phi, \quad R\Phi = -2c\Phi + cq(x) + cq(y). \end{aligned}$$

All contractions and particle martingales use the already proved fixed- N domain. In particular $B\Phi$ has absolutely integrable Haar slices by its structural equation. This does not follow from an absolute bound on $|K| |\nabla \Phi|$.

Averaged trace and lower contraction. The inverse sup bound and its common-translation covariance give, for every multi-index α ,

$$\|\Phi_t[h]\|_\infty \leq C\sqrt{N} \|h\|_{C^2}, \quad (\partial_x + \partial_y)^\alpha \Phi_t[h] = \Phi_t[\partial^\alpha h].$$

Difference quotients converge in the off-diagonal sup norm; individual relative derivatives need not have such a uniform bound. Put $\psi_x(z) = \Phi_t(x, x - z)$ and $D = \nabla_x + \nabla_y$. Then

$$B\Phi_t(x, x - z) = K(z) \cdot D\Phi_t(x, x - z) + 2K(z) \cdot \nabla_z \psi_x(z).$$

The first term is integrable because $K \in L^1$; the second scalar product is integrable by subtraction from the legitimate contraction. On the punctured torus the inner normal is $-\theta$ and $K(\varepsilon\theta) \cdot (-\theta)\varepsilon^3 = -2 + O(\varepsilon^4)$. Thus

$$\int_{|z|>\varepsilon} K \cdot \nabla \psi_x = c \int_{|z|>\varepsilon} \psi_x - 2 \int_{\mathbb{S}^3} \psi_x(\varepsilon\theta) dS + O(\varepsilon^4 \|\Phi_t\|_\infty).$$

The two volume integrals converge, proving existence of the normalized spherical average $\tau_t(x)$ and $|\tau_t(x)| \leq \|\Phi_t\|_\infty$. No directionwise or pointwise diagonal value is assigned. With $b_t(x) = \int B\Phi_t(x, y) dy$ and $c = 2|\mathbb{S}^3|$, the exact identity is

$$b_t(x) = \int K(z) \cdot D\Phi_t(x, x - z) dz + 2c(q_t(x) - \tau_t(x)).$$

Both the negative spherical flux and positive periodic compensation remain. It follows that $\|b_t[h]\|_\infty \leq C\sqrt{N} \|h\|_{C^3}$. The bounded linear map $h \mapsto b_t[h]$ is translation covariant, so difference quotients yield

$$\sup_t \|b_t[h]/\sqrt{N}\|_{C^j} \leq C\|h\|_{C^{j+3}}, \quad \int b_t[h] = b_t[\int h] = 0.$$

The last equality uses that a constant terminal test has zero source. It is not true for every arbitrary symmetric pair kernel.

Actual lower-drift estimate. For $\rho_t = \eta_N(t) - dx$, the actual energy inequality and positive heat split give

$$\mathbb{E}|\widehat{\eta}_N(t, k)|^2 \leq \min(1, CN^{-1/2}|k|^2), \quad k \neq 0.$$

Consequently every deterministic smooth test a , including an N, t, ν_N dependent one, satisfies

$$\mathbb{E}|\rho_t[a]| \leq CN^{-1/4} \sum_{k \neq 0} |k| |\widehat{a}(k)| \leq CN^{-1/4} \|a\|_{C^6}.$$

The last series converges in dimension four using $(1 - \Delta)^3$. Therefore the entire lower drift

$$\ell_N(t) = \frac{1}{N} \rho_t[b_t] + \frac{1}{2N} \int b_t$$

obeys $\sqrt{N} \mathbb{E} \int_0^T |\ell_N(t)| dt \leq CN^{-1/4}$. This is an absolute estimate under the actual evolved law; it does not replace that law by a product measure.

Exact source identity and controlled noise. The literal deleted statistic is

$$P_N[v] = \frac{1}{2N^2} \sum_{i \neq j} v(X_i, X_j) - \frac{1}{N} \sum_i \int v(X_i, y) dy + \frac{1}{2} \iint v.$$

All U_3 backgrounds have their original coefficients. If $C\Phi = \text{Sym}_3[K(x - z) \cdot \nabla_x \Phi(x, y)]$, with the six-permutation average, the exact domain identity gives

$$\int_0^T P_N[J_t] dt = P_N[\Phi_0] + \int_0^T U_3[C\Phi_t] dt + \int_0^T \ell_N(t) dt + M_T^2[\Phi],$$

$$M_t^2[v] = \sqrt{2\nu_N} \sum_i \int_0^t \nabla_i P_N[v_a](X_a) \cdot dW_i(a).$$

The iid initial endpoint has $\mathbb{E}|\sqrt{N}P_N[\Phi_0]| \leq C\sqrt{(1 + \log N)/N}$, retaining the small first projection and mean before bounding them.

Let $\Psi_{t,\delta} = e^{\delta\Delta_x} e^{\delta\Delta_y} \Phi_t$. Its uniform Haar L^1 bound and heat Fourier sums imply

$$Q_N[\Psi_\delta] := 2\nu_N N \mathbb{E} \int_0^T \sum_i |\nabla_i P_N[\Psi_{t,\delta}]|^2 dt \leq C\nu_N (N^{-1/2}\delta^{-10} + N^{-2}\delta^{-9}).$$

The second term retains the missing empirical self label. With $\delta_N = N^{-1/40}$ and $\nu_N = O(N^{-1/2})$, this gives $Q_N[\Psi_{\delta_N}] \leq CN^{-3/4}$ and $\mathbb{E}|\sqrt{N}M_T^2[\Psi_{\delta_N}]| \leq CN^{-3/8}$.

Set $V_N = \Phi - \Psi_{\delta_N}$, $S_N = \sqrt{N} \int_0^T P_N[J_t] dt$ and

$$W_N = \sqrt{N} \left[\int_0^T U_3[C\Phi_t] dt + M_T^2[V_N] \right].$$

Both martingales are genuine at each finite N , and

$$\mathbb{E}|S_N - W_N| \leq C \left(N^{-1/4} + \sqrt{\frac{1 + \log N}{N}} + N^{-3/8} \right) \rightarrow 0.$$

Thus the original assertion is equivalent to $\mathbb{E}|W_N| \rightarrow 0$, which remains unproved. The cubic and residual martingale are retained together; their separate decay would be sufficient, but is not asserted necessary. The finite initial segment with $\min(\beta_N, 1) \neq 1$ changes no limit. This result gives no Gaussian law, static-law counterexample or hierarchy closure.